

# Time & Work — Fast Tricks & Shortcuts

LCM Method, Worker-Day Tricks, and Every Speed Technique You Need

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This file collects the **fast calculation methods** — the actual shortcuts you execute under exam pressure. For the full catalogue of question *types* that use these tricks, see `03-question-pattern-catalog.tex`.

## 1 The LCM Method (Best Trick in the Topic)

Instead of taking total work = 1 and juggling fractions, take total work = **LCM of all the given times**. Every individual's rate then becomes a clean whole number.

### Worked Example

Individual times = 10 days, 15 days, 30 days.

Total work =  $\text{LCM}(10, 15, 30) = 30$  units

$A$ 's rate =  $30/10 = 3$  units/day

$B$ 's rate =  $30/15 = 2$  units/day

$C$ 's rate =  $30/30 = 1$  unit/day

No fractions anywhere — just add rates directly.

This is the single fastest technique in the whole topic and should be your default approach whenever every given time is an integer.

### Why This Beats the Fraction Method

Fractions with different denominators need a common denominator anyway before you can add them — the LCM method just does that step *once, up front*, and keeps every number that follows as a clean integer.

## 2 Rate-First Thinking for Ratio Questions

Whenever a question gives an efficiency or time **ratio** instead of concrete numbers, assign rates immediately instead of trying to imagine days.

### Rule

“ $A$  is  $k$  times as efficient as  $B$ ”  $\rightarrow$  let  $B$ 's rate = 1 unit/day,  $A$ 's rate =  $k$  units/day. Never start by guessing how many days each person takes.

This was introduced in `01-fundamentals.tex` §6.1 — it’s repeated here because it is, by a wide margin, the highest-leverage mental habit in this topic. It also combines naturally with the LCM method: once you have a rate ratio, picking total work = LCM of the *implied* times keeps everything as clean integers.

### 3 Worker-Day Method — Workers Leaving

Use the conserved quantity Total Work = Workers  $\times$  Days (from `01-fundamentals.tex` §10) directly, rather than switching to fractional one-day’s-work.

#### Procedure

1. Total work (worker-days) = Original Workers  $\times$  Original Days
2. Work done so far = Workers present  $\times$  Days already worked
3. Remaining work = Total work – Work done so far
4. Days needed = Remaining work / Workers now present

#### Worked Example

24 workers can finish a job in 18 days. After 6 days, 6 workers leave. How many more days to finish?

$$\begin{aligned}\text{Total work} &= 24 \times 18 = 432 \text{ worker-days} \\ \text{Done in 6 days} &= 24 \times 6 = 144 \text{ worker-days} \\ \text{Remaining} &= 432 - 144 = 288 \text{ worker-days} \\ \text{Workers left} &= 24 - 6 = 18 \\ \text{Days needed} &= 288/18 = 16 \text{ more days}\end{aligned}$$

### 4 Worker-Day Method — Workers Joining Later

Exactly the mirror image of §3: compute the work already completed by the original group, subtract it from the total, then divide the remaining work by the **new**, larger workforce.

#### General Procedure

1. Compute Total Work as either 1 (fractional method) or worker-days (whole-number method).
2. Compute work already done before the new workers joined.
3. Solve only for the **remaining** work with the **updated** rate.

This “completed work, then remaining work” split is the master pattern behind almost every midway-change question — joining, leaving, or a mix of both.

## 5 Alternate-Day / Cyclic Working — Compute the Full Cycle First

When two (or more) workers alternate days (A on day 1, B on day 2, A on day 3, ...), **don't** calculate day-by-day from scratch. Instead:

### Procedure

1. Calculate the work done in **one full cycle** (usually 2 days, or however long the repeating pattern is).
2. Divide the total work by the cycle work to find how many **complete cycles** fit.
3. Handle the **leftover** work with the next partial day(s) separately.

### Worked Example

A's one-day work =  $1/10$ , B's one-day work =  $1/15$ . They alternate, starting with A.

$$\text{2-day cycle work} = \frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$$

Since  $1/6$  of the work finishes every 2 days, the whole job needs  $6 \times 2 = 12$  days — and because  $1/6 \times 6 = 1$  exactly, there's no partial final day to handle here.

### Common Mistake

Assuming the two workers contribute equally per cycle. They don't — always use each individual's actual one-day rate inside the cycle, and pay attention to **whose turn is last** if the work finishes mid-cycle.

## 6 Efficiency Change → Time Change (Percentage Method)

Efficiency and time are inversely proportional, so a percentage change in efficiency does **not** translate into the same percentage change in time.

### Formulas

If efficiency increases by  $x\%$ :

$$\text{New Time} = \text{Old Time} \times \frac{100}{100 + x}$$

If efficiency decreases by  $x\%$ :

$$\text{New Time} = \text{Old Time} \times \frac{100}{100 - x}$$

**Worked Example — Increase**

Efficiency +25%, old time = 20 days.

$$\text{New Time} = 20 \times \frac{100}{125} = 20 \times \frac{4}{5} = 16 \text{ days}$$

**Worked Example — Decrease**

Efficiency -20%, old time = 15 days.

$$\text{New Time} = 15 \times \frac{100}{80} = 15 \times \frac{5}{4} = 18.75 \text{ days}$$

**Common Mistake**

Computing “efficiency +25%  $\Rightarrow$  time -25%.” Time drops by **less** than the efficiency gain, because the relationship is a reciprocal, not linear.

## 7 Wages — Proportional to Work Share

Restating 01-fundamentals.tex §11 as an executable procedure:

**Procedure**

1. Find each person’s individual rate (or use the ratio directly).
2. Split the total wage in the same ratio as the rates — **not** the time each person spent working.

**Worked Example**

A (10 days) and B (15 days) work together and jointly earn Rs. 1500.

$$\text{Rate ratio } A : B = 1/10 : 1/15 = 3 : 2 \quad (\text{multiply both by 30})$$

$$A\text{'s share} = 1500 \times 3/5 = \text{Rs. } 900$$

$$B\text{'s share} = 1500 \times 2/5 = \text{Rs. } 600$$

## 8 Pipes and Cisterns — Handling a Pipe That’s Closed Partway

A very common TCS-style twist: both pipes open together, but one is closed after some time.

**Procedure**

1. Compute the net rate while **both** pipes are open, and find how much work that portion completes.
2. Subtract from 1 to get the **remaining** work.

3. Switch to the rate of whichever pipe is **still running** and solve only for the remainder.

### Worked Example

Fill pipe A = 10 hr, outlet pipe B = 15 hr. Both open together, but B (the outlet) is closed after 3 hours. Find the total time to fill the tank.

$$\text{Net rate (both open)} = 1/10 - 1/15 = 1/30$$

$$\text{Work in first 3 hr} = 3 \times 1/30 = 1/10$$

$$\text{Remaining work} = 1 - 1/10 = 9/10$$

$$\text{A alone (rate } 1/10) \text{ needs: } (9/10)/(1/10) = 9 \text{ more hours}$$

$$\text{Total time} = 3 + 9 = 12 \text{ hours}$$

## 9 Quick Reference — Which Trick for Which Situation

### Lookup Table

| Situation                                 | Use                              |
|-------------------------------------------|----------------------------------|
| All given times are clean integers        | LCM Method (§1)                  |
| A ratio is given instead of numbers       | Rate-First Thinking (§2)         |
| Workers leave partway through             | Worker-Day Method (§3)           |
| Workers join partway through              | Worker-Day Method, mirrored (§4) |
| Two/more workers alternate days           | Cycle-first method (§5)          |
| Efficiency changes by a percentage        | Percentage-to-time formula (§6)  |
| Payment needs to be split among workers   | Wages-by-work-share (§7)         |
| One pipe is closed/opened partway through | Split into phases (§8)           |